TECHNICAL SPECIFICATION · V2.0

OXOT Cyber Digital Twin

The OXOT Seldon Engine is a physics-based Cyber Digital Twin for critical infrastructure. It fuses facility physics, Purdue topology, plant engineering data and external intelligence into one deterministic model, then translates cyber pathways into financial loss exposure. Architecture is inside-out plant engineering fusion — safety, reliability, operations and device cascades — grounded outside-in by actuarial and threat data.

CLASS	Technical & product reference, v2.0	AUDIENCE	CISO · plant manager · chief architect · risk actuary · M&A lead
ENGINE	Seldon — physics-based risk and prediction stack	DEPLOYMENT	Single-tenant island, EU sovereign cloud or on-premises

01 · SEVEN-LAYER ARCHITECTURE STACK

L	LAYER	SPECIFICATION
07	**Governance**	ALE financial exposure engine · Consequence Index · Monte Carlo alternate history · auto-generated compliance technical files
06	**Services**	AI/ML process optimisation · interactive P&ID canvas · Analyst Studio visualisers · Red Squadron adversary emulation
05	**Data fusion**	Inside-out plant engineering fusion (FMECA · SCIL · RCIL · MOR) · device cascade modelling · outside-in actuarial and conflict feeds
04	**Networks**	Virtual network state · VLAN, subnet and virtual firewall modelling · Purdue L0–L4 segmentation and DMZ verification · PCAP flow analysis
03	**Interoperation**	P&ID NER/NLP extraction pipeline into DEXPI 2.0 · OPC UA, MQTT, Modbus TCP, DNP3, EtherNet/IP, PROFINET, BACnet parsing · CycloneDX integration
02	**Assets**	PLC ladder logic and structured text parsing · SCADA, RTU and HMI configuration mapping · field instruments, actuators and virtualised elements
01	**Facility physics**	Thermodynamics and fluid dynamics · process kinetics (pipe burst, valve failure, thermal runaway) · environmental and containment limits

MODEL HIERARCHY

- **Five drill levels** — component → equipment → line → facility → organization, traversable in both directions
- **Five projections** — P&ID, Purdue, network, dependency graph and 3D site view of one object
- **Synchronisation** — continuous; BOMs, risk deltas and file sections regenerate as differences

STANDARDS EMBEDDED IN THE MODEL

DEXPI 2.0	CycloneDX
IEC 62443 zones	TS 50701
Purdue L0–L4	MITRE ATT&CK

02 · INSIDE-OUT GROUNDING

Plant engineering integration.

Consequence is taken from the organisation's own engineering record rather than estimated at the security layer. Three operational teams supply the ground truth, and the twin binds cyber pathways to the loss figures those teams have already signed off.

INPUT	STANDARD	WHAT IT CONTRIBUTES
FMECA register	IEC 60812	Failure mode, effects and criticality per equipment item
Hazard log	ISO 31000	Centralised hazard tracking scored on a 5×5 risk matrix
SCIL	SIF · SIL 1–4	Safety-critical controllers, valves and sensors mapped to their instrumented functions
Worst-case mapping	TS 50701	Cyber pathways linked to physical HAZOP scenarios
RCIL	Reliability	Assets whose failure degrades reliability or causes component fatigue
RAMS analysis	EN 50126	Reliability, availability, maintainability and safety across production lines
Device cascades	Purdue L2→L0	How compromise of one PLC tag propagates into mechanical failure downstream
MOR thresholds	Operations	Minimum operating requirements to keep the plant running safely
Downtime curves	Financial	Loss per hour of lost production, per product line — the basis of every euro figure

INGESTION PIPELINE

- **P&ID NER/NLP** — un-annotated CAD and legacy PDF drawings parsed for equipment tags, line numbers, valve types and interlocks
- **Schema** — DEXPI 2.0 XML extended with CycloneDX data models
- **Control logic** — ladder logic, structured text, SCADA/RTU/HMI configuration
- **Network** — topology exports and PCAP flow data for boundary inspection

INDUSTRIAL PROTOCOL COVERAGE

OPC UA	Modbus TCP
DNP3	EtherNet/IP
PROFINET	BACnet
MQTT	TCP/IP

CONSEQUENCE CHAIN

01 · Tag compromised
A reachable pathway terminates at a specific PLC tag or field device.

02 · Cascade traced
Device cascade modelling propagates the failure across equipment and plant boundaries.

03 · Physics evaluated
Layer 01 determines whether the result is a trip, a spill, a burst or a runaway.

04 · Loss attached
SCIL, RCIL and MOR supply severity; downtime curves convert it to currency.

03 · THREAT ACTOR INTELLIGENCE

TACAM and ATQ.

TACAM is a pre-computed knowledge graph profiling threat groups across seven independent dimensions; ATQ scores each actor's capability, motivation and sector convergence. Cross-dimensional queries resolve in under a second — for example, which actors target energy using Modbus TCP against a named PLC family, with the CVE pathways and kill-chain coverage attached.

TTP cluster Preferred ATT&CK tactics	**Sector affinity** 17 CISA sectors	**Geography** Origin and target regions	**Protocol capability** OT protocols exploitable
Temporal rhythm Tempo, recency, dormancy	**CPE match** Vendor products targeted	**CWE concentration** Weakness families used	**ATQ score** 12-factor, 0–100

04 · WORLDMONITOR EXTERNAL PRESSURE

News & geopolitics	Military & security
Maritime & aviation	Cyber & infrastructure
Climate & disasters	Energy & commodities
Finance & economy	Government & supply chain
Nuclear & strategic facilities	31-Country Instability Index

OUTSIDE-IN DATA SOURCES

SEC EDGAR 8-K disclosed incidents
NetDiligence claims benchmarks
Breach-investigation frequency studies
World Bank governance indicators
ACLED conflict event data
Every value stored with its citation

05 · FIVE BILLS OF MATERIALS · CYCLONEDX

SBOM	Software and firmware, with transitive dependency tracing through library depth
HBOM	Microcontrollers, ASICs, PLC backplanes and hardware origin risk
CBOM	Key lengths, certificate expiry, cipher suites and post-quantum readiness
SaaS-BOM	Cloud APIs, vendor maintenance tunnels and external service dependencies
Ops-BOM	Workflows, maintenance schedules and human access roles

VULNERABILITY ENRICHMENT

KEV — known exploited	**EPSS** — exploit probability
CVSS — severity in context	**CAPEC** — attack patterns

Exploitability is scored as a reachable pathway through the modelled topology.

REPRESENTATIVE QUERIES

Which actors target our sector using our vendors' products?
What threat exposure do we inherit if we adopt a new vendor?
Which actors are converging on our sector this quarter?
Which reachable pathways terminate at a safety-critical function?

06 · SIMULATION AND ACTUARIAL ENGINE

Seldon SLT	Loss/likelihood transformer forecasting severity and likelihood across the modelled graph
Severity fitting	Log-normal fitting of empirical breach losses from disclosed filings and claims studies
Tail modelling	Conditional value at risk at the 95th and 99th percentiles for catastrophic events
Monte Carlo	MITRE-aligned attack campaigns run to convergence, returning breach probability with confidence intervals
Alternate history	Random walks over accidental change, configuration drift and proposed investment
ALE output	Annual loss expectancy in EUR/USD, marrying internal MOR downtime curves to external claims

07 · PRIORITISATION AND SCORING

NOW / NEXT / NEVER — triage by consequence × exploitability	**Consequence Index** — one board-level number, 90-day trend
Glass box — every figure drills to the component tag and source filing	**Supplier rating** — CVE, CWE, EPSS, CAPEC and ATT&CK history curves

12 · OUTPUTS AND INTERFACES

Machine-readable — CycloneDX BOMs and DEXPI 2.0 model exports	**Board reporting** — Consequence Index, ALE and tail exposure on trend
Engineering views — interactive P&ID canvas, network, Purdue, graph and 3D	**Regulatory files** — generated technical files per framework, with evidence links

08 · COMPLIANCE OUTPUTS

EU CRA Annex VII	IEC 62443	NIS2	TS 50701
NERC CIP	TSA directives	CFATS	AI Act · Machine Act

13 · ANALYST STUDIO AND ADVERSARY EMULATION

3D threat globe — attack arcs from origin regions to your sites	**Radial threat radar** — proximity-based actor ranking
Kill-chain viewer — multi-stage attack flow mapping	**Briefing studio** — automated briefings from platform research
Red Squadron AI — adversary emulation generating realistic campaigns against the twin, never against the plant	

09 · DEPLOYMENT OPTIONS

OPTION 1	OPTION 2	OPTION 3
Island mode Isolated, on your own ground. No external dependencies, no access to control systems, and your own custom AI model.	**One-way data diode** A data diode limits traffic to inbound only. Our intelligence streams into the Cyber Digital Twin, and nothing exits.	**Dedicated instance** A single-tenant instance in an AWS service of your choice, aligned to your data sovereignty requirements.

All three are passive-first: no agents on PLCs, RTUs or controllers, and no active scanning of the process network.

10 · INTEGRATIONS

Asset management	Historians	Network monitoring	Service management

14 · DATA PROVENANCE AND QUALITY DISCIPLINE

Grounding first — real source data retrieved before any value is synthesised	**No fabrication** — a value that cannot be tied to a real disclosure is omitted
Null over zero — unsourced fields stay empty so completeness stays honest	**Citations stored** — every figure carries the source it derived from

11 · ENGAGEMENT MODELS

OXOT OT engineering consulting builds and deploys the Cyber Digital Twin against your environment. Our collaborartive onboarding process builds the model with your teams.

- **Transient** — a one-time consulting engagement, model built, deliverables generated and torn down.
- **Long-term operations** — the twin runs on alongside your teams, with key capabilities available..

One environment. One evolving model. Better security decisions.

Bring one P&ID and an asset list for a single facility and we will build the model against this specification.

Contact Us @ info@oxot.nl

The Consequence Index, ALE and simulations are OXOT's own transparent, drillable calculations, directionally validated on real engagements — not rating-agency or actuarial marks. Specification reflects v2.0 and is subject to change.